

1 Introduction

Here I want to formulate the motivation for the paper.

2 Literature Review

Financial markets exhibit periods of stress, increased volatility and negative returns . Investors try to mitigate the risks arising from these movements, by diversifying their investments internationally Solnik (1974). However, in times of high volatility markets, empirical evidence suggests that correlations between international markets increase Ang and Bekaert (2002). This is especially relevant to international investors, because the benefits of diversification are lost.

The market conditions underlying such periods are difficult to measure because they are not directly observable. Instead, they must be inferred from the often times very noisy financial data. Hamilton (1989) introduced a Markov-switching framework for modeling unobserved changes in the dynamics of macroeconomic business-cycles. The same framework has subsequently been adapted to financial markets , where latent states are interpreted as market regimes characterized by different conditional return distributions, volatilities, and covariance structures.

Within the relevant literature, it is proposed to account for market regimes, in the context of portfolio construction Ang and Bekaert (2002). Typical indications for market regimes that can be classified as "bear" markets for example, include often periods of increased volatility, increased international market correlations or a decrease in expected returns.

There is also empirical evidence that financial markets exhibit nonstationarity, meaning that return distributions are not constant and change over time. In an analysis on the dynamics of the S&P 500, Stărică and Granger (2005) for example report that returns are primarily concentrated in shifts of the unconditional variance, suggesting that nonstationarities significantly affect the behavior of stock returns. Additionally, the findings also raise questions around persistence in stock returns, meaning whether past events influence current return dynamics, or if the observed nonstationarities are the primary drivers of persistence.

References

- Ang, A. and Bekaert, G. (2002). International asset allocation with regime shifts. *The Review of Financial Studies*, 15(4):1137–1187.
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384.
- Solnik, B. H. (1974). Why not diversify internationally rather than domestically? *Financial Analysts Journal*, 30(4):48–54.
- Stărică, C. and Granger, C. (2005). Nonstationarities in stock returns. *The Review of Economics and Statistics*, 87(3):503–522.